Start coding or generate with AI.

```
import pandas as pd

# Create a custom dataset
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva', 'Frank','merry'],
    'Age': [25, 30, 35, 40, 22, 29,26],
    'Department': ['HR', 'Engineering', 'Engineering', 'Sales', 'HR', 'Sales','HR'],
    'Salary': [50000, 80000, 85000, 70000, 48000, 72000,90000],
    'Years_Exp': [2, 7, 10, 15, 1, 4,8]
}

df = pd.DataFrame(data)
df
```

	Name	Age	Department	Salary	Years_Exp
0	Alice	25	HR	50000	2
1	Bob	30	Engineering	80000	7
2	Charlie	35	Engineering	85000	10
3	David	40	Sales	70000	15
4	Eva	22	HR	48000	1
5	Frank	29	Sales	72000	4
6	merry	26	HR	90000	8

Next steps: Generate code with df New interactive sheet

```
head = df.head(3)
head
```

	Name	Age	Department	Salary	Years_Exp
0	Alice	25	HR	50000	2
1	Bob	30	Engineering	80000	7
2	Charlie	35	Engineering	85000	10

Next steps: Generate code with head New interactive sheet

```
tail = df.tail(2)
tail
```

	Name	Age	Department	Salary	Years_Exp
5	Frank	29	Sales	72000	4
6	merry	26	HR	90000	8

Next steps: Generate code with tail New interactive sheet

```
shape = df.shape
shape
```

```
(7, 5)
```

```
col = df.columns
col
```

```
Index(['Name', 'Age', 'Department', 'Salary', 'Years_Exp'], dtype='object')
```

```
tyspe = df.dtypes
tyspe
```

	0
Name	object
Age	int64
Department	object
Salary	int64
Years_Exp	int64

dtype: object

```
col1 = df["Name"]
col1
```

	Name
0	Alice
1	Bob
2	Charlie
3	David
4	Eva
5	Frank
6	merry

dtype: object

```
df_sorted = df.sort_values(by='Salary', ascending=False)
print("Employees sorted by Salary:")
display(df_sorted)

# Create a new column 'Is_Senior' if Years_Exp > 5
df['Is_Senior'] = df['Years_Exp'] > 5
print("\nDataFrame with Seniority column:")
display(df)
```

```
locl1 = df.loc[1]
locl1
```

	1
Name	Bob
Age	30
Department	Engineering
Salary	80000
Years_Exp	7

dtype: object

```
short2 = df.sort_values("Age")
short2
```

	Name	Age	Department	Salary	Years_Exp
4	Eva	22	HR	48000	1
0	Alice	25	HR	50000	2
6	merry	26	HR	90000	8
5	Frank	29	Sales	72000	4
1	Bob	30	Engineering	80000	7
2	Charlie	35	Engineering	85000	10
3	David	40	Sales	70000	15

Next steps: Generate code with short2 New interactive sheet

```
f1 = df[df["Salary"] > 50000]
f1
```

	Name	Age	Department	Salary	Years_Exp
1	Bob	30	Engineering	80000	7
2	Charlie	35	Engineering	85000	10
3	David	40	Sales	70000	15
5	Frank	29	Sales	72000	4
6	merry	26	HR	90000	8

Next steps: Generate code with f1 New interactive sheet

```
df['rating'] = [3,5,3,2,4,1,5]
df
```

	Name	Age	Department	Salary	Years_Exp	rating
0	Alice	25	HR	50000	2	3
1	Bob	30	Engineering	80000	7	5
2	Charlie	35	Engineering	85000	10	3
3	David	40	Sales	70000	15	2
4	Eva	22	HR	48000	1	4
5	Frank	29	Sales	72000	4	1
6	merry	26	HR	90000	8	5

Next steps: Generate code with df New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Name        7 non-null      object
 1   Age         7 non-null      int64
 2   Department  7 non-null      object
 3   Salary      7 non-null      int64
 4   Years_Exp   7 non-null      int64
 5   rating      7 non-null      int64
dtypes: int64(4), object(2)
memory usage: 468.0+ bytes
```

```
df.describe()
```

	Age	Salary	Years_Exp	rating
count	7.000000	7.000000	7.000000	7.000000
mean	29.571429	70714.285714	6.714286	3.285714
std	6.187545	16377.975920	4.889250	1.496026
min	22.000000	48000.000000	1.000000	1.000000
25%	25.500000	60000.000000	3.000000	2.500000
50%	29.000000	72000.000000	7.000000	3.000000
75%	32.500000	82500.000000	9.000000	4.500000
max	40.000000	90000.000000	15.000000	5.000000

```
df = df.drop(df[df['Age'] > 29].index)
display(df)
```

	Name	Age	Department	Salary	Years_Exp	rating
0	Alice	25	HR	50000	2	3
4	Eva	22	HR	48000	1	4
5	Frank	29	Sales	72000	4	1
6	merry	26	HR	90000	8	5

Next steps: Generate code with df New interactive sheet

```
ex = df["Salary"].max()
ex
```

```
90000
```